

Section A — Concept Application

S1 Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

I would normally start with the CPU and then choose the other parts around it. First, I would identify the CPU socket and supported chipset. The motherboard must have the correct socket and support the processor generation; BIOS/UEFI support must also be checked because a matching socket alone does not guarantee CPU support.

I would select RAM that matches the motherboard and CPU requirements, including the correct DDR generation, supported capacity, number of DIMMs and supported speed. The motherboard manual and, where available, its Qualified Vendor List (QVL) should be checked. Intel recommends checking processor memory specifications and the motherboard vendor's support information.

For storage, I would check whether the board provides the required SATA ports or compatible M.2 slots and whether the intended M.2 drive uses a supported NVMe/PCIe interface. I would also verify case form factor, physical clearance, PSU connectors and total power requirements.

Before buying the parts, I would check the motherboard support list, RAM details, storage interface and PSU requirements to make sure everything matches. A test boot should confirm successful POST and detection of the CPU, RAM and storage in BIOS/UEFI.

S2 Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

An HDD stores data on rotating magnetic platters and uses moving read/write heads. It normally offers large capacity at relatively low cost but has higher access latency, mechanical noise and lower random-access performance. A SATA SSD uses flash memory with no moving parts and connects through SATA, making it much faster and quieter than an HDD.

NVMe is a storage protocol designed for non-volatile memory and is commonly used with PCIe-connected M.2 SSDs. It provides higher bandwidth and lower latency than SATA because it uses PCIe. Crucial's reference material notes that SATA SSDs are commonly around the 500 MB/s range, while NVMe SSDs can be several times faster depending on the drive and PCIe generation.

For a normal office PC, I would choose an SSD. If the motherboard supports NVMe, I would use an NVMe SSD as the main drive. Office users benefit from faster boot times, quicker application loading, responsive file access and quiet operation. HDDs remain useful for inexpensive bulk storage, but an SSD is the better primary system drive.

S3 Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

The GPU performs graphics-processing tasks such as rendering images, interfaces and 2D/3D workloads. When software supports GPU acceleration, a suitable GPU can reduce graphics-processing work on the CPU and improve responsiveness.

During several hours of sustained use, the CPU and GPU generate heat. The cooling system removes this heat through the CPU cooler, GPU cooler, case fans, heatsinks and airflow paths. Good thermal contact,

correct thermal paste where required, unobstructed airflow and clean filters help maintain safe temperatures.

If temperatures become too high, components can reduce their operating frequency through thermal throttling, lowering performance. Excessive heat can also cause instability or crashes. So, good cooling is important because it helps the PC keep its performance steady during long periods of use.

S4 Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Compatibility is important because the parts need to fit together physically and also support each other electrically and technically. The CPU must match the motherboard socket and supported chipset/BIOS; RAM must match the supported DDR type and platform limits; storage must use a supported interface; the GPU must fit and have the required power connections; and the PSU must provide sufficient wattage and connectors.

Intel states that processor sockets are not interchangeable and that motherboard BIOS support is also required for compatibility.

Possible problems include failure to POST, RAM not being detected, storage not appearing in BIOS/UEFI, instability, insufficient power, incorrect connectors, physical mounting problems or components that cannot be installed because of incompatible dimensions or interfaces.

S5 Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

I would list the CPU, GPU, motherboard, RAM, storage, cooling, fans and peripherals, then estimate their full-load power requirements. I would use a reputable PSU calculator as a cross-check and choose a quality PSU with sufficient continuous wattage and reasonable headroom for transient loads and future upgrades. Seasonic and Corsair both recommend considering the complete system and allowing additional headroom.

Before powering on, I would switch the PSU off and disconnect AC power while checking connections. I would verify the 24-pin ATX motherboard connection, CPU EPS/ATX12V connection, GPU power connection and SATA power connections. I would also confirm that modular cables are specifically compatible with the PSU.

I would inspect for loose screws, exposed conductors, damaged connectors, trapped cables, incorrect motherboard standoffs and possible shorts. GPU power connectors should be fully seated. Corsair specifically advises proper seating of PCIe power connectors and use of the manufacturer's supplied cables.

S6 Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

I would power on the system and observe whether it starts normally. A successful POST should allow the system to reach BIOS/UEFI or begin the operating-system boot process without a hardware error. I would check POST LEDs, beep codes or a diagnostic display if the motherboard provides them.

In BIOS/UEFI, I would verify that the CPU is identified correctly, the expected RAM capacity is detected, the storage device is listed and temperatures are reasonable. I would also confirm the correct boot device.

If POST fails, I would then check the hardware step by step instead of changing several things at once. With AC power removed where appropriate, I would check the 24-pin and CPU power connections, reseal RAM, verify CPU installation and cooling, check GPU seating/power, inspect for shorts, and test with minimum hardware. I would use the motherboard's diagnostic indicators and manual to narrow the fault.

Section B — Practical Tasks

Task 1 Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system. Note the make/model or key specification of each component where visible. Present the findings as a labeled component list or table.

Suggested component identification table:

Component	Function	Observed Make/Model	Key Specification
Motherboard	Main circuit board connecting CPU, RAM, storage, expansion cards, power and I/O.	ASUS PRIME B250-PLUS	LGA1151 / Intel B250 / ATX
CPU	Executes instructions and performs general-purpose processing.	Intel Core i5-7500	4 cores / up to 3.8 GHz / LGA1151
RAM	Temporary working memory used by the CPU and applications.	Kingston 8 GB DDR4	8 GB / DDR4-2400
GPU	Processes graphics and display workloads.	NVIDIA GeForce GTX 1060 6 GB	6 GB GDDR5 / PCIe
Storage device	Persistent storage for the OS, applications and files.	Samsung 850 EVO 250 GB	250 GB / SATA SSD
Cooling system	Transfers and removes heat to maintain safe operating temperatures.	Intel stock CPU cooler	Air cooler / fan-based

Task 2 Identify and label all major motherboard ports, expansion slots, and the CPU socket. Identify the CPU socket type and note which CPU families are compatible with it.

When looking at the motherboard, I would identify the following ports and connectors, where they are present: CPU socket; DIMM/RAM slots; 24-pin ATX motherboard power; CPU EPS/ATX12V power; SATA ports; M.2 slots; PCIe x16 and other PCIe slots; internal USB headers; front-panel header; front-panel audio; fan headers; RGB/ARGB headers; CMOS battery; rear USB ports; Ethernet; audio jacks; display outputs; and diagnostic LEDs/buttons.

Record the exact socket label from the motherboard or its manual, then use the motherboard manufacturer's CPU support list. Intel confirms that sockets must match and BIOS support is also required.

Task 3 Safely remove and reinstall a RAM module and a storage device, documenting ESD precautions, correct seating, connection type, and detection after reinstallation.

- Shut down the PC, switch off the PSU, disconnect AC power and use an ESD-safe work area. Handle RAM and storage by their edges and avoid touching contacts.
- Release the RAM retaining clips, remove the module, note the slot used, then reinstall it with the notch aligned to the slot key until the retaining clips lock.
- For SATA storage, disconnect SATA data and power. For an M.2 NVMe drive, remove the retaining screw, slide the drive out, then reinstall it into the keyed slot and secure it.
- Boot to BIOS/UEFI and confirm the expected RAM capacity and storage device are detected. Intel recommends checking memory slot configuration and compatibility against the motherboard manual/QVL.
- If a component is missing, power down safely and re-check seating, slot choice, cables, power and firmware settings before replacing hardware.

Task 4 Assemble a desktop PC from the provided motherboard, CPU, RAM, storage, GPU and PSU; follow the correct installation sequence; document the steps; power on; and confirm POST.

Recommended installation sequence:

- Prepare the workspace, use ESD precautions and verify all parts against the build list.
- Install the CPU using the socket's alignment/keying markers. Do not force it. Intel's installation guidance stresses correct alignment and seating before securing the loading mechanism.
- Install the CPU cooler and connect its fan/pump to the correct motherboard header.
- Install RAM in the recommended DIMM slots and confirm the retaining clips are engaged.
- Install the M.2 NVMe drive, if used, or prepare the SATA storage drive and cables.
- Install the motherboard in the case using the correct standoffs and screws.
- Install the PSU and connect the 24-pin motherboard and CPU EPS power connectors.
- Install the GPU in the correct PCIe slot, secure it and connect its required power cable.
- Connect SATA power/data, front-panel switches/LEDs, USB headers, audio and case fans as applicable.
- Perform a final visual inspection for loose screws, unsecured connectors, trapped cables and incorrect connections.
- Power on, confirm POST, enter BIOS/UEFI and verify CPU, RAM, storage, temperatures and boot device.

Section C — Mini Capstone Project

Capstone 1 Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

The first three areas I would check are: (1) PSU/power connections, (2) RAM, and (3) CPU/motherboard compatibility and installation. A missing power connection can prevent startup; incorrectly seated or incompatible RAM can prevent POST; and an unsupported CPU, incorrect socket installation or motherboard/BIOS problem can stop the platform before normal boot.

I would also use POST LEDs, beep codes or a diagnostic display if available. If the PSU appears functional but the system still fails, I would reduce the system to minimum POST hardware and test one variable at a time.

Capstone 2 Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

The CPU must be mechanically and electrically compatible with the motherboard socket, and the chipset and BIOS/UEFI must support the CPU generation. An incorrect socket or unsupported CPU can cause no POST even when the CPU appears to fit. Intel states that sockets are not interchangeable and that BIOS support is also required.

For the inspected system, record: motherboard model = ASUS PRIME B250-PLUS, CPU model = Intel Core i5-7500, socket = LGA1151, chipset = Intel B250, BIOS version = [enter the BIOS version shown on the inspected system]. Compare the CPU with the motherboard manufacturer's support list. This avoids inventing hardware details that were not supplied.

Capstone 3 Explain how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

For RAM, verify that each module is fully seated and the notch matches the slot. Use the recommended DIMM slots, then enter BIOS/UEFI and confirm the expected total memory capacity and configuration are detected.

For storage, verify the physical connection: M.2 drive fully inserted and secured, or SATA drive connected to both SATA data and power. BIOS/UEFI should list the expected SATA or NVMe device in storage information.

If BIOS/UEFI does not detect a component, power down safely and check seating, slot choice, cables, power, compatibility and firmware settings before assuming the component is defective.

Capstone 4 Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.

- Compatibility check before assembly: CPU socket/chipset/BIOS, RAM type/capacity, storage interface, case fit, GPU clearance and PSU wattage/connectors.
- ESD and physical inspection: use ESD precautions, inspect connectors and verify motherboard standoffs and mounting points.

- Connection verification: confirm 24-pin ATX, CPU EPS, GPU power, storage power/data, RAM seating, CPU cooler connection and front-panel wiring.
- POST verification: enter BIOS/UEFI and confirm CPU, RAM, storage, temperatures and boot device.
- Documentation: record component models, BIOS version, test results and photos/screenshots.

Capstone 5 Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

SSDs are preferred as primary storage because they have no moving mechanical parts, provide faster access to data, reduce boot and application-loading times, operate quietly and generally use less power than HDDs. Crucial describes SSDs as faster and more power efficient because they have no moving parts.

For the ABC Solutions lab, faster application launch and file access improve responsiveness across office workloads. An NVMe SSD is especially suitable where the motherboard supports it because it uses PCIe and can provide substantially higher throughput than SATA. HDDs can still be used for low-cost bulk or archival storage.

Section D — AI-Augmented Learning

Step 1 — Build with AI Use an AI tool to produce a step-by-step troubleshooting checklist for a desktop PC that fails to boot immediately after assembly, covering motherboard, RAM, storage and PSU, and apply it to the ABC Solutions scenario.

Exact prompt used:

“Create a clear, on-site troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the checks in the correct priority order. Cover at minimum the motherboard, RAM, storage, and PSU as possible failure points. Start with ESD and physical safety. Include what the technician should check, what a normal result looks like, and what to do if the check fails. Apply the checklist to the ABC Solutions scenario in which some newly assembled PCs fail to start while others show hardware-detection errors. Keep the checklist practical and suitable for a CompTIA A+ hardware assessment.”

AI Original Checklist AI's original checklist.

- 1. Make sure the PC is connected to power and the PSU switch is on.
- 2. Check that the motherboard 24-pin power connector and CPU power connector are connected.
- 3. Check the RAM modules are seated correctly and try one module at a time if necessary.
- 4. Check the CPU is installed correctly and the CPU cooler is connected.
- 5. Check the GPU is seated and powered if a dedicated GPU is required.
- 6. Check the motherboard for diagnostic LEDs, beep codes or error messages.
- 7. Enter BIOS/UEFI and check whether RAM and storage are detected.
- 8. Check SATA cables or M.2 installation if storage is missing.
- 9. Verify the boot drive is selected correctly in BIOS/UEFI.
- 10. If the problem remains, test the PSU or replace it with a known-good compatible unit.
- 11. Record the fault and the component that resolved it.

Step 2 — Test & Debug Corrected checklist after testing/debugging the AI version.

- 1. SAFETY FIRST — Shut down, switch the PSU off, disconnect AC power when opening/reseating components, and use appropriate ESD precautions.
- 2. Confirm the external power path: wall power/AC cable, PSU switch and obvious signs of PSU operation. Do not open the PSU enclosure.
- 3. Check motherboard installation for incorrect standoffs, loose screws, visible damage or shorts. Confirm 24-pin ATX and CPU EPS/ATX12V connectors are fully seated.
- 4. Check CPU installation and cooling: correct socket alignment, locking mechanism, cooler mounting and CPU-fan/pump connection.
- 5. Check RAM: correct type, recommended slot, full seating and locking clips. Test with minimum RAM configuration and one known-good module where appropriate.
- 6. Check GPU and display path if required: correct PCIe slot, secure seating, required power and monitor cable connected to the correct video output.
- 7. Use motherboard POST diagnostics: LEDs, beep codes or diagnostic display. Consult the motherboard manual for the code meaning.

- 8. Enter BIOS/UEFI if POST is successful. Confirm CPU identification, RAM capacity, storage detection, CPU temperature and boot order.
- 9. If storage is missing, power down safely and verify M.2/SATA installation, SATA data/power, M.2 slot compatibility and BIOS/UEFI storage settings.
- 10. If the system still does not POST, verify CPU–motherboard compatibility and BIOS support, then test with minimum hardware and known-good compatible components.
- 11. Test the PSU with an appropriate tester or known-good compatible PSU if required. Use only compatible manufacturer-approved modular cables.
- 12. Document the symptom, diagnostic evidence, failed check, corrective action and final POST/BIOS result.

ABC Solutions application

The corrected checklist first separates complete no-start/no-POST systems from systems that power on but show hardware-detection errors. No-POST systems should prioritize power, motherboard, CPU and RAM. Systems reaching BIOS but missing hardware should then prioritize RAM/storage detection and physical connections. This prevents software-level troubleshooting before the physical platform is verified

Debug Note 3–4 line note explaining what was changed and why the AI version needed it.

I moved ESD and physical safety to the first step because components should not be reseated while the system is energized. I added motherboard-standoff/short checks and made CPU–motherboard/BIOS compatibility explicit because a system can fail POST even when power is present. I also separated POST diagnosis from BIOS/storage diagnosis so physical boot-critical hardware is checked before configuration or operating-system issues.